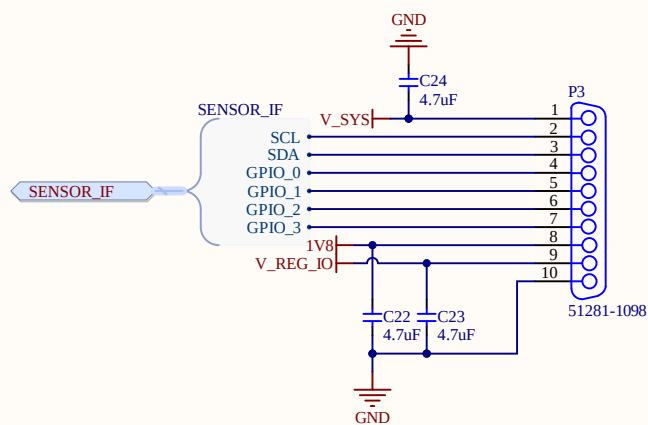




		OpenRing-V1R0		
Designer/Engineer Yusein Ali		Board Title ORing-RevA		
Drawn By Yusein Ali		Document Name OpenRing_DaughterBoard.SchDoc		
Approved By Dariush Salami		Rev. A0	Sheet 9 of 9	Date 01.05.2025

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A

A

B

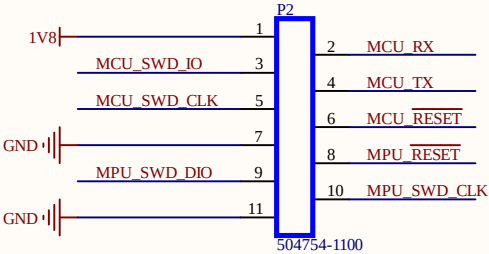
B

C

C

D

D



OpenRing-V1R0

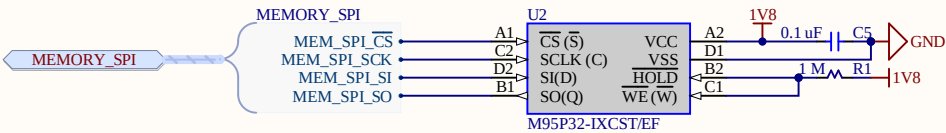
Designer/Engineer	Yusein Ali	Board Title	ORing-RevA
Drawn By	Yusein Ali	Document Name	OpenRing_Debug.SchDoc
Approved By	Dariush Salami	Rev.	A0
		Sheet	7 of 9
		Date	01.05.2025
		Size	A4

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		OpenRing-V1R0	
Designer/Engineer Yusein Ali		Board Title ORing-RevA	
Drawn By Yusein Ali		Document Name OpenRing_Memory.SchDoc	
Approved By Dariush Salami		Rev. A0	Sheet 3 of 9 Date 01.05.2025 Size A4

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POWER

POWER

CHARGER_INT

PWR_KILL

GAUGE_INT

REG_EN

PWR_I2C_SCL

PWR_I2C_SDA

CHG_IRQ

PWR_KILL

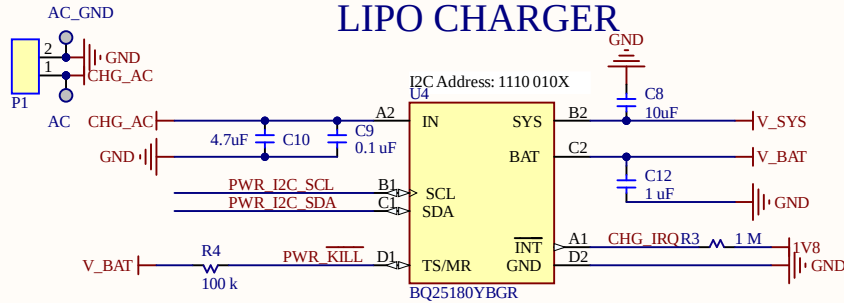
GAUGE_IRQ

V_REG_EN

PWR_I2C_SCL

PWR_I2C_SDA

LIPO CHARGER

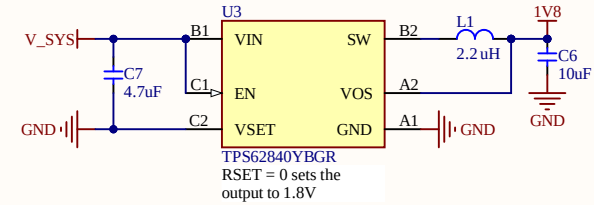


NTC related parameters can be calculated using External NTC Monitoring calculator GUI available at https://dev.ti.com/gallery/view/BatteryChargers/ticharger_gui

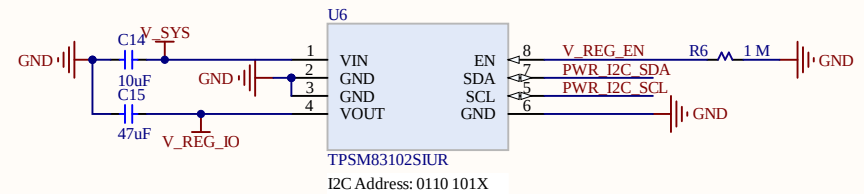
$R_p = 1\text{ M}$ is required to allow different beta values of the NTC

The selected NTC has beta value of 3380

1. @PWR_KILL/A can be used as long press based shutdown signal if it is needed.
2. CHG_AC must be connected to USB VBUS.
3. V_BAT is the supply from GAUGE, not BATTERY pack + terminal.
4. NTC must be placed to the bottom of the board to indicate something about the case temperature.
5. BATTERY must have another NTC for GAUGE to operate properly.

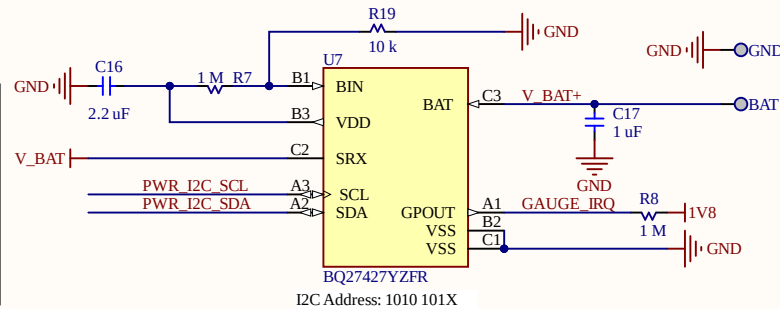


TPS62840YBGR
RSET = 0 sets the output to 1.8V



TPSM83102SIUR
I2C Address: 0110 101X

LIPO GAUGE



BATTERY PACK must have

- VBAT+
- VBAT-
- BAT_DET pins.

BAT_DET pin is connected to thermistor as below.



OpenRing-V1R0

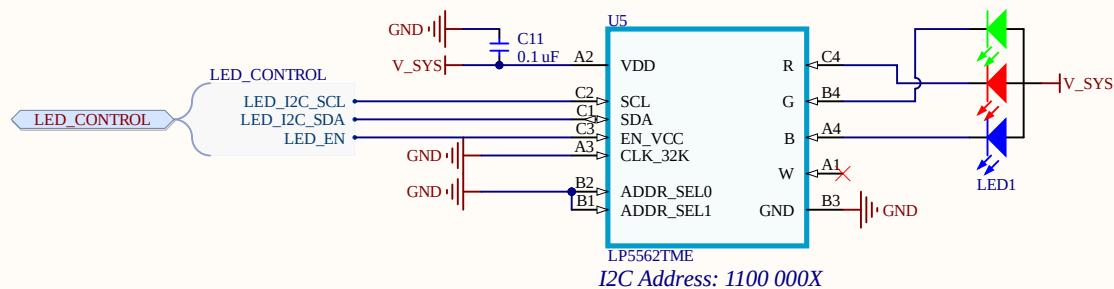
Designer/Engineer	Yusein Ali	Board Title	ORing-RevA
Drawn By	Yusein Ali	Document Name	OpenRing_Power.SchDoc
Approved By	Darius Salami	Rev.	A0
Sheet	4	of	9
Date	01.05.2025	Size	A4

1

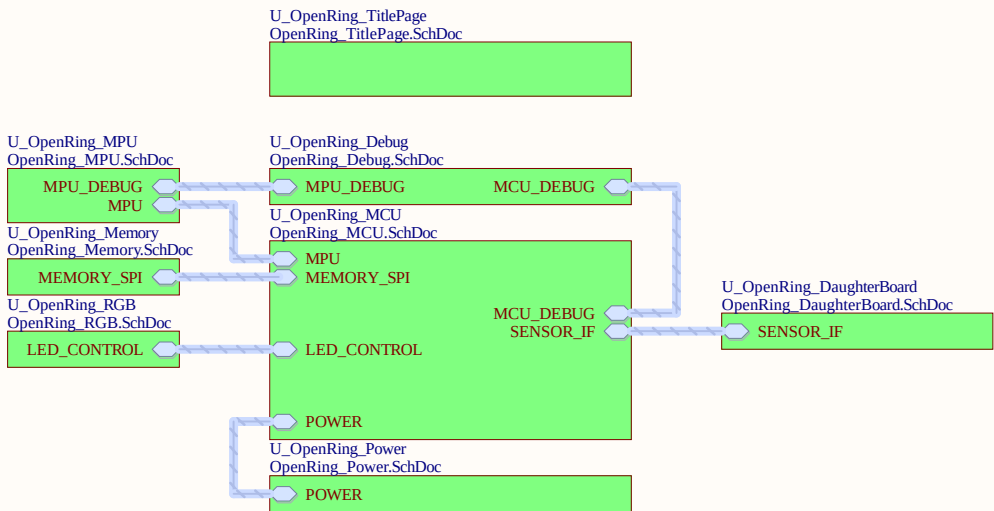
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		OpenRing-V1R0		
Designer/Engineer		Board Title		
Yusein Ali		ORing-RevA		
Drawn By		Document Name		
Yusein Ali		OpenRing_RGB.SchDoc		
Approved By		Rev.	Sheet	Size
Dariush Salami		A0	6 of 9	A4
		Date	01.05.2025	



		OpenRing-V1R0		
Designer/Engineer Yusein Ali		Board Title ORing-RevA		
Drawn By Yusein Ali		Document Name OpenRing-Main.SchDoc		
Approved By Dariush Salami		Rev. A0	Sheet Date	1 of 9 01.05.2025 Size A4